

Data Analytics – Course Report

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1 Statistics

Choosing the right statistical method depends on whether the data supports the normality assumption.

question	parametric	nonparametric
comparing two groups	t-test	Mann-Whitney U
comparing 3+ groups	ANOVA	Kruskal-Wallis
relationship of two continuous variables	Pearson correlation / regression	Spearman correlation
association of two categorical variables	Chi-square	Fisher's exact

1.1 Two-groups Comparison (t-test)

When we run a t -test, the sample data is condensed into a single number, the t -statistic, which measures how many standard errors the observed result sits away from what H_0 predicts. To turn that number into a p -value, R places it on a t -distribution – a bell-shaped curve whose exact shape is governed by the degrees of freedom, $df = n - 1$. For small df this curve has heavier tails than the normal distribution, reflecting the extra uncertainty that comes from estimating variance off a small sample; as df grows, the t -distribution converges to the standard normal. The p -value is then simply the area under this curve beyond the observed $|t|$ (both tails, for a two-sided test), as shown in Figure 1: a larger $|t|$ falls further into the tail, leaving less area beyond it, and hence a smaller p -value. Historically, this area was found by comparing t against a critical value read off a printed table for the chosen significance level α and df – for a two-sided test at $\alpha = 0.05$, for instance, t needs to fall at or beyond the 97.5th percentile of the distribution (in absolute value), since the 5% is split evenly across the two tails. Software now performs this same area calculation directly, which is why R reports a p -value rather than requiring a table lookup.

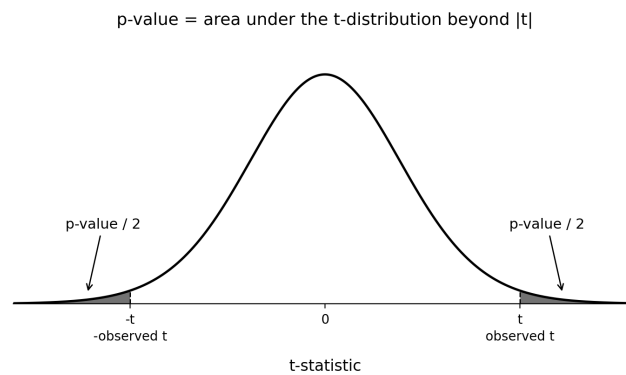


Figure 1: The p -value as the area under the t -distribution beyond the observed test statistic. The shaded regions in both tails together sum to the p -value; a larger $|t|$ pushes further into the tail, leaving less area beyond it, i.e., a smaller p -value.